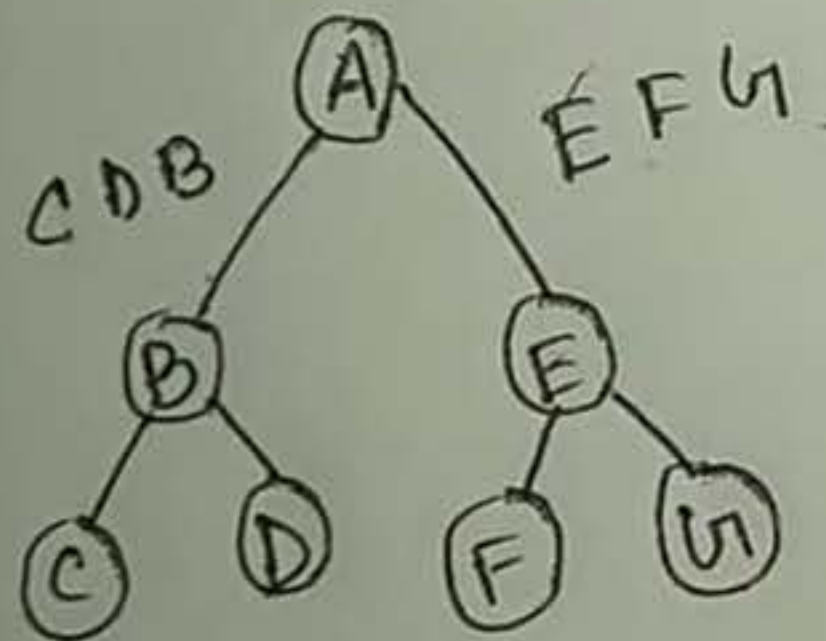


Example 1:

Made with KINEMASTER

preorder: A, B, C D E F G [Root, L, R]

Post order: C D B F G E A [L, R, Root]



pre: B C D

post: C D B

pre: E F G

post: F G E

✓

Construct Binary Search Tree from Preorder and Postorder

- ① Preorder $\rightarrow$ Left side
 $\Rightarrow$ Left side (elements) $\rightarrow$ Postorder
- ② Postorder $\rightarrow$ Right side
 $\Rightarrow$ Right side (elements) $\rightarrow$ Preorder.

Exception!!

Note: Left side Root element hai, jiska right element hai.

Preorder: 10, 6, 4, 3, 5, 8, 18, 15, 21, 22 [Root, L, R]

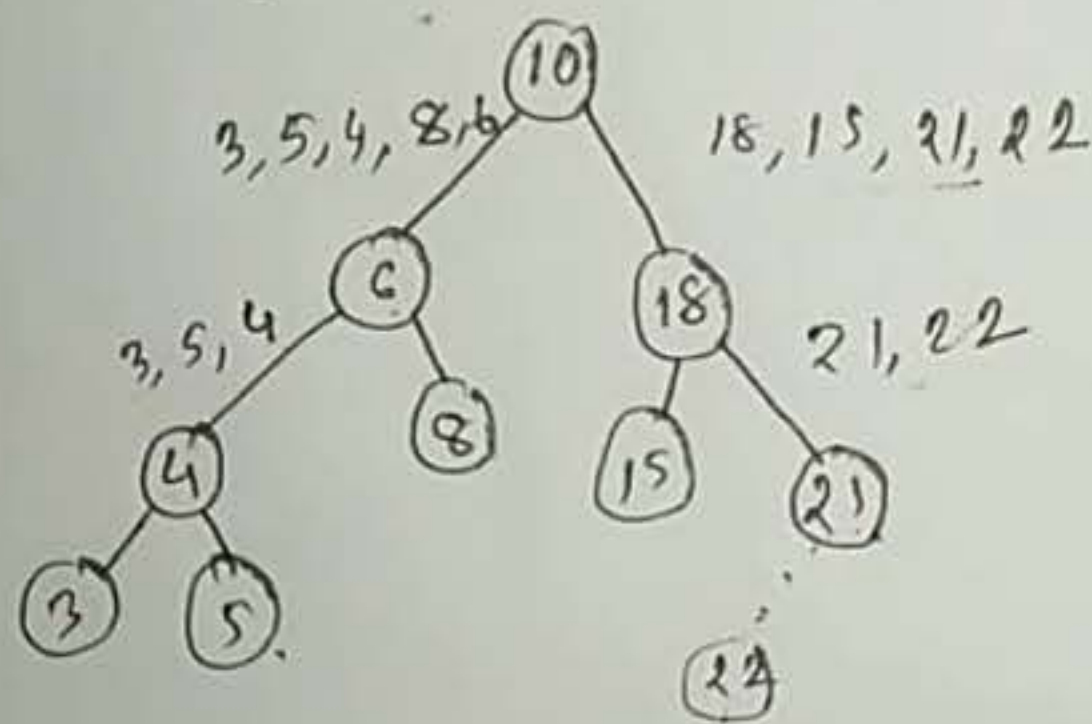
Postorder: 3, 5, 4, 8, 6, 15, 22, 21, 18, 10 [L, R, Root]

Note: Left side (elements) $\rightarrow$ Preorder.

Left side Root right side $\rightarrow$ Right side root

Preorder: 10, 6, 4, 3, 5, 8, 18, 15, 21, 22 [Root, L, R]

Postorder: 3, 5, 4, 8, 6, 15, 22, 21, 18, 10 [L, R, Root]



Pre: 6, 4, 3, 5, 8

Post: 3, 5, 4, 8, 6

Pre: 4, 3, 5

Post: 3, 5, 4

Pre: 18, 15, 21, 22

Post: 15, 22, 21, 18

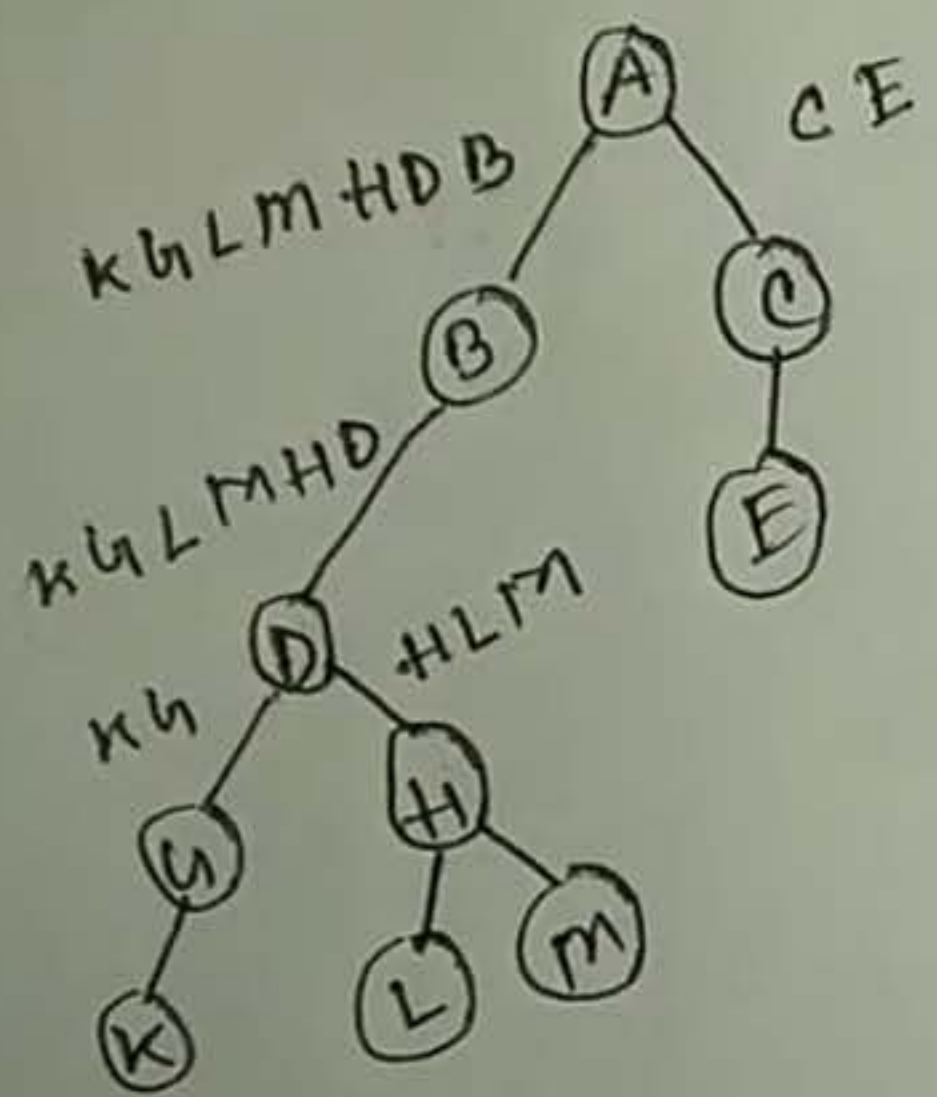
Pre: 21, 22

Post: 22, 21

Exception!!

22 can be left or right child according to the sequence, but since $21 < 22$ so 22 cannot be a left child

postorder: K G L M H D B E C A [L, B, K, G, M, H, D, B, E, C, A]



pre → B D G K H L M
 post → K G L M H D B

pre → D G K H L M
 post: K G L M H D

pre: G K
 post: K G

pre: H L M
 post: L M H

pre: C E
 post: E C

Root, L, R]

L, R, Root]

K H L M

M H D B

K H L M

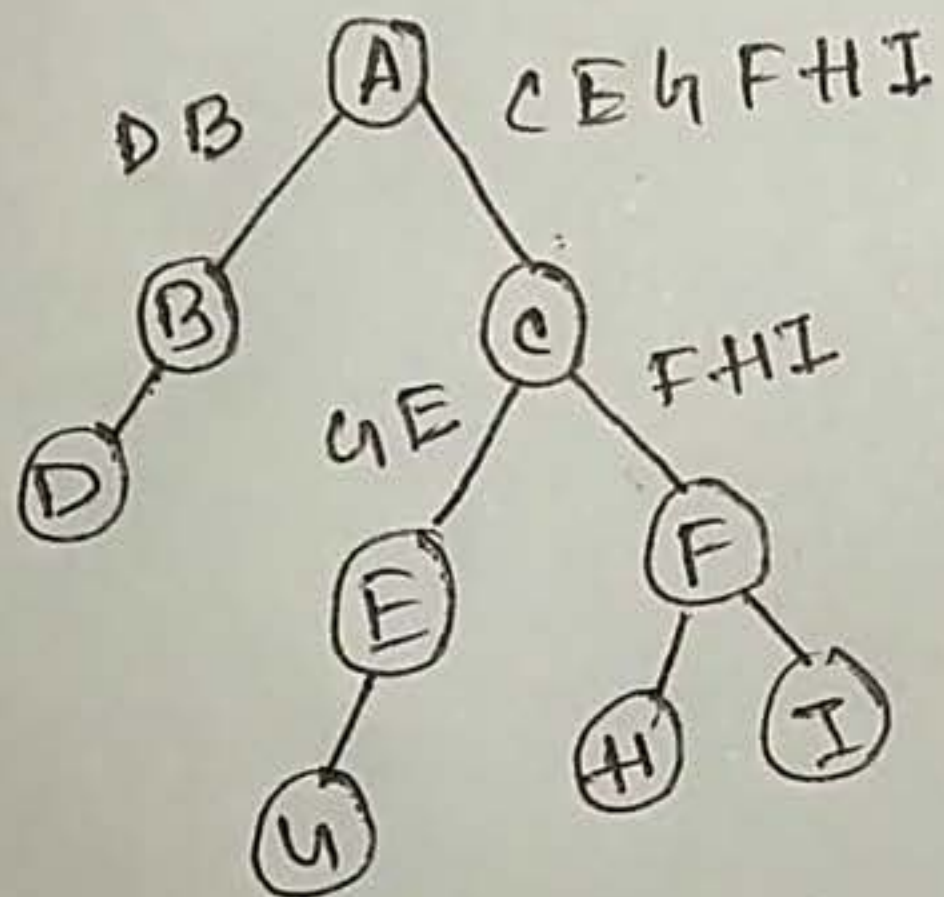
L M H D

L M

M H

Preorder: A B D C E G F H I [^{Root, L, R} ~~L, R, Root~~]

Postorder: D B G E H I F C A [L, R, Root]



Pre: B D
Post: D B

Pre: C E G F H I
Post: G E H I F C

Pre: E G
Post: G E

Pre: F H I
Post: H I F

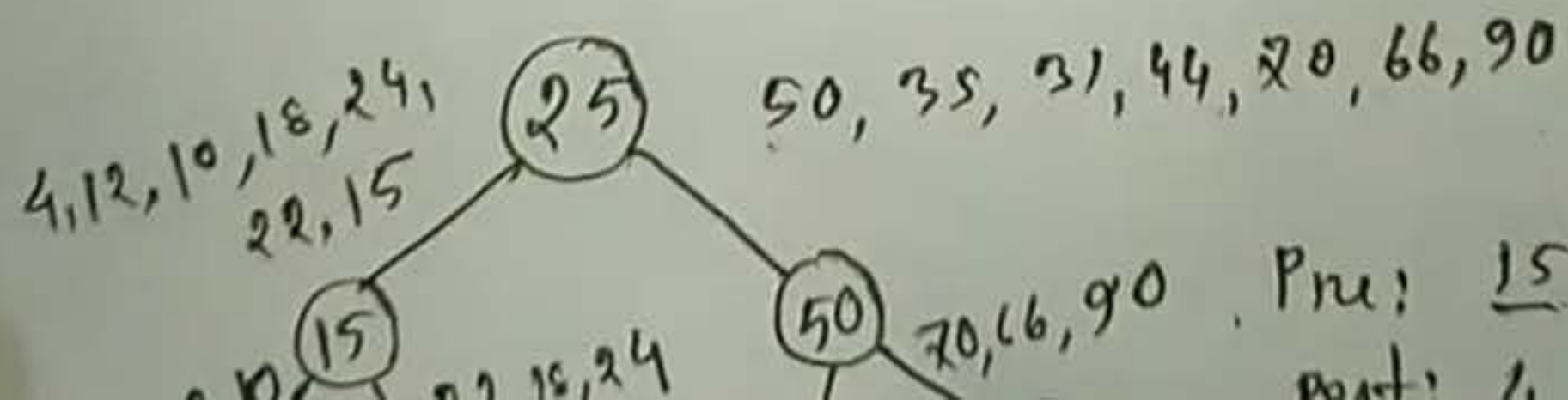
i) Preorder $\rightarrow$ Left side
 $\Rightarrow$ Left side (element) $\rightarrow$ Postorder

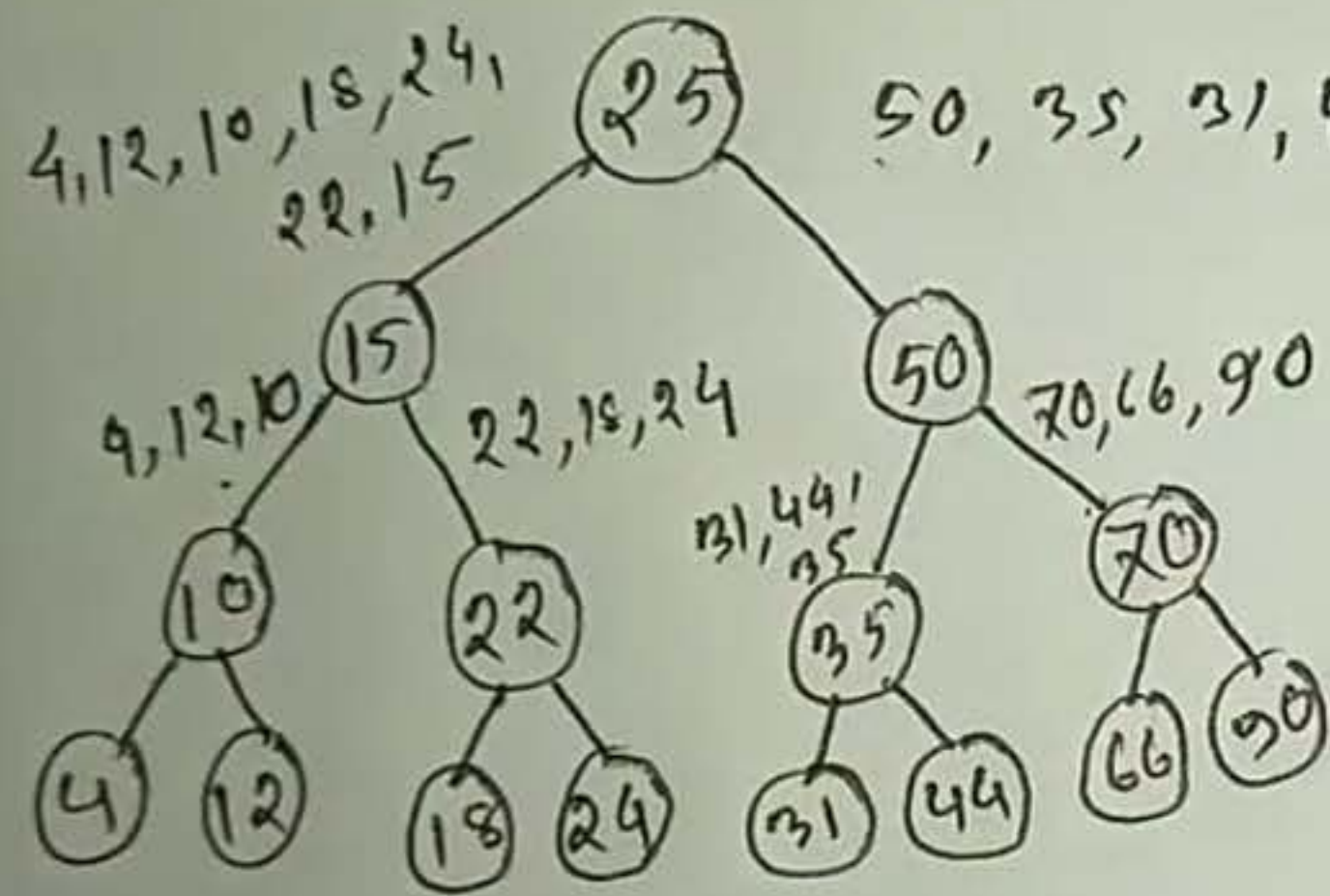
ii) Postorder $\rightarrow$ Right side
 $\Rightarrow$ Right side (element) $\rightarrow$ Preorder.

Example: 4

preorder: 25, 15, 10, 4, 12, 22, 18, 24, 50, 35, 31, 44, 70, 66, 90 [Root, L, R]

postorder: 4, 12, 10, 18, 24, 22, 15, 31, 44, 35, 66, 90, 70, 50, 25 [L, R, Root]





Pre: 15, 10, 4, 12, 22, 18, 24
 Post: 4, 12, 10, 18, 24, 22, 15

Pre: 10, 4, 12 | Pre: 22, 18, 24
 Post: 4, 12, 10 | Post: 18, 24, 22

Pre: 50, 35, 31, 44, 70, 66, 90
 Post: 31, 44, 35, 66, 90, 70, 50

Pre: 35, 31, 44 | Pre: 70, 66, 90
 Post: 31, 44, 35 | Post: 66, 90, 70